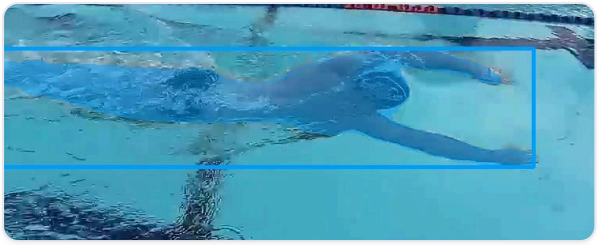
Swimline swim video analysis

Drop in a phone video of a swim, get splits, speed, strokes and coaching notes

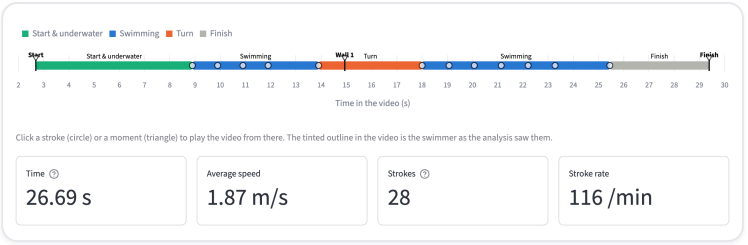
PYTHON · STREAMLIT · SAM 2

Personal project · research, models and app, built solo



Swimmer tracking

SAM 2 outline and box, found and re-found automatically



Results page

Clickable race timeline: start, strokes, turn, finish

Project overview

Swimline turns **an ordinary phone video of a freestyle swim into race numbers with no manual input**: splits, speed, stroke count, stroke rate, distance per stroke, turn times, SWOLF and stroke index, plus suggestions and what-if projections that quote the numbers behind them.

Off-the-shelf pose models fail in splash, so the pipeline uses only signals the video shows reliably: a segmentation track of the swimmer, the rhythm of the arm lifting out of the water, the camera's pan, and the start beep in the audio. Every metric was checked against hand counts, an on-screen stroke counter and official race results.

VISION	SAM 2 · YOLO-World · OpenCV
SIGNAL	Autocorrelation · Optical flow · Audio FFT · NumPy
APP	Streamlit · Altair · pandas
MEDIA	PyAV · H.264 / libx264
RUNTIME	PyTorch · Apple MPS · CUDA
TESTING	Ground-truth evaluation · Playwright

Engineering challenges

01 Finding a swimmer in splash

PROBLEM

A stock person detector found the swimmer in 0–3 of 40 frames, and pose models scored ~0.35 confidence on every joint. A panning camera and other lanes made it worse.

SOLUTION

Open-vocabulary YOLO-World ("swimmer") seeds SAM 2 mask tracking. Masks that jump, balloon or turn into a lane rope are rejected, and the tracker is re-seeded from the detector path whenever it loses or strays from the swimmer.

IMPACT

Swimmer followed in 92–100% of frames on side-view clips, with no manual marking.

02 Counting strokes without a skeleton

PROBLEM

Without reliable joints there is no wrist to follow, and splash hides strokes and adds false ones.

SOLUTION

How far the mask reaches above its own centre rises with each arm recovery. Autocorrelation finds the cycle; strokes count only where the signal is locally periodic, and strokes hidden by splash are filled in from the rhythm.

IMPACT

Stroke rate within 1% of an on-screen counter (56.2 vs 56/min); counts within about 2 strokes.

03 Timing from a moving camera

PROBLEM

The camera pans with the swimmer, so the swimmer barely moves in the picture, and walls are rarely in view.

SOLUTION

The camera's pan (median optical flow outside the swimmer) is subtracted from the swimmer's motion: a direction flip is a wall. The start comes from the electronic beep, a steady ~1 kHz tone found by FFT, and the finish from the last stroke.

IMPACT

Turn found within 0.3 s; on an NCAA broadcast, the beep put official splits on the exact frames shown.

Measured against ground truth

±1%	stroke rate against an on-screen stroke counter
0.3 s	turn time error, found automatically
~5%	50 m total time against the official result
16 vs 15	automatic stroke count against a hand count

Responsibilities

Solo project 100%

- Research & model selection
- CV pipeline
- Signal processing
- Evaluation
- UI / UX
- Data visualization

PROJECT HIGHLIGHTS

- ✓ Zero-input analysis
- ✓ SAM 2 segmentation tracking
- ✓ Open-vocabulary detection
- ✓ Pose-free stroke counting
- ✓ Optical-flow timing
- ✓ Audio start detection
- ✓ Interactive race timeline
- ✓ Coaching notes & what-ifs